Logic definitions

1 Explain what the following terms mean in Logic:

a Inductive reasoning
b Deductive reasoning
c Statement
d Simple statement
e Compound statement

2 State whether the following words are quantifiers:

a All
b Some
c Not
d None

3 A restaurant dinner menu states:
"All dinners are served with a choice of salad or soup, and potatoes or pasta, and carrots or mushrooms."
Here 'or' is the exclusive or. It means one or the other, but not both. Give an example of a selection that is allowed.

Compound statements and negation

4 For each of the following statements:

i State whether it is a simple or compound statement.
ii If it is a compound statement, state what kind of compound statement it is.

a If you write a biography, then the life of Nelson Mandela would be a good choice.
b Mr Jones is teaching Statistics during the spring semester.
c Time will go forward if, and only if, you travel faster than the speed of sound.
d Oliver did not finish his homework.
e Emma joined the Army and she got injured.
f The man is neither African nor European.
g If Lachlan plays in the World Cup, then he will play for Tonga.
h The cyclone did $\$600\,000$ worth of damage to Rockdale.
i Georgia will be admitted to law school if, and only if, she completes her undergraduate degree.

5 Write down the negation of the following statements:

a All humans are able to swim.

b All houses are connected to the internet.

c Some birds do not have flippers.

d No Librans are Capricorns.

e No tricycles have two wheels.

f All cats have feet.

g Some cars are driving across the bridge.

h Some doctors are not paediatricians.

i No skydivers are scientists.

j Some biscuits contain sugar.

Symbolic form

6 Write down the symbol that corresponds to the following:

a Negation
b Conjunction
c Disjunction
d Conditional
e Biconditional

7 Consider the statement: "Vincent started the fire and Valerie did not forget the meat".

a Write the statement in symbolic form by letting:

- p: Vincent started the fire.
- q: Valerie forgot the meat.

b What type of statement is this?

8 Consider the statement: "If the meeting is in Sydney, then we can see a film or we can go to the concert".

a Write the statement in symbolic form by letting:

- p: The meeting is in Sydney.
- q: We can see a film.
- r: We can go to the concert.

b What type of statement is this?

9 Consider the statement: "It is false that if you eat healthy foods, then you will not gain fat".

a Write the statement in symbolic form by letting:

- p: You eat healthy foods.
- q: You gain fat.

b What type of statement is this?

10 Consider the statement: "If lunch is ready then we can eat, or we cannot go to the restaurant".

a Write the statement in symbolic form by letting:

- p: Lunch is ready.
- q: We can eat.
- r: We can go to the restaurant.

b What type of statement is this?

11 Consider the statement: "If the food has salt or the food has fat, then you will like it".

a Write the statement in symbolic form by letting:

- p: The food has salt.
- q: The food has fat.
- r: You will like the food.

b What type of statement is this?

12 Consider the statement: "If Yvonne is the instructor then Joanne is in Science class, if and only if, it is not a public holiday."

a Write the statement in symbolic form by letting:

- p: Yvonne is the instructor.
- q: Joanne is in Science class.
- r: It is a public holiday.

b What type of statement is this?

13 Consider the statement: "You may enrol in [redacted]eometry II class if, and only if, you did not fail Geometry I or you passed the placement exam."

a Write the statement in symbolic form by letting:

- p: You may enrol in this Geometry II class.
- q: You failed Geometry I.
- r: You passed the placement exam.

b What type of statement is this?

14 Consider the statement: "If the motorbike has fuel and the battery has power, then the motorbike will start".

a Write the statement in symbolic form by letting:

- p: The motorbike has fuel.
- q: The battery has power.
- r: The motorbike will start.

b What type of statement is this?

15 Consider the statement: "The cinema is closed if, and only if, it is a public holiday, or it is before 7 am.

a Write the statement in symbolic form by letting:

- p: The cinema is closed.
- q: It is a public holiday.
- r: It is before 7 am.

b What type of statement is this?

16 Let

- p: A zebra has a mane.
- q: A dog can bark.

Write the following statements in symbolic form:

a A zebra does not have a mane.

b A zebra has a mane and a dog can bark.

c A dog cannot bark or a zebra does not have a mane.

d A dog cannot bark if, and only if, a zebra does not have a mane.

e If a zebra does not have a mane, then a dog cannot bark.

f A dog cannot bark, but a zebra has a mane.

17 Let

- p: The lemonade is sour.
- q: The rice is hot.

Write the following statements in symbolic form:

a The rice is not hot if, and only if, the lemonade is sour.

b If the lemonade is sour, then the rice is not hot.

c The lemonade is not sour, but the rice is hot.

d Neither is the lemonade sour nor is the rice hot.

e It is false that the lemonade is sour or the rice is hot.

f It is false that if the rice is not hot, then the lemonade is sour.

18 Let

- p: Dave has a laptop.
- q: James has a smartphone.

Write the following statements in words:

a $\sim q$ **b** $p \wedge q$ **c** $\sim p \implies q$ **d** $\sim p \vee \sim q$

e $\sim (p \wedge q)$

19 Let

- p: Quentin has a rabbit.
- q: Maximilian has a dog.

Write the following statements in words:

a $\sim p$ **b** $p \vee q$ **c** $\sim p \iff \sim q$ **d** $\sim (p \vee q)$

e $\sim p \wedge \sim q$

20 Let

- p: The temperature is 90°.
- q: The heater is working.
- r: The apartment is cold.

Write the following statements in symbolic form:

a The temperature is 90° and the heater is not working, and the apartment is cold.

b The temperature is not 90° and the heater is working, but the apartment is cold.

c The temperature is 90° and the heater is working, or the apartment is cold.

d If the temperature is 90°, then the heater is working or the apartment is not cold.

21 Let

- p: The temperature is 80°.
- q: The fan is working.
- r: The apartment is hot.

Write the following statements in symbolic form:

a If the apartment is hot and the fan is working, then the temperature is 80°.

b The temperature is not 80° if, and only if, the fan is not working, or the apartment is not hot.

c The apartment is hot if, and only if, the fan is working, and the temperature is 80°.

d If the fan is working, then the temperature is 80°, if and only if, the apartment is hot.

22 Let

- p: The water is $28^{\circ}\mathrm{C}$.
- q: The sun is down.
- r: We go swimming.

Write the following statements in words:

a $(p \vee q) \wedge \sim r$ **b** $\sim p \wedge (q \vee r)$ **c** $\sim r \implies (q \wedge p)$ **d** $(q \implies r) \wedge p$

e $(q \iff p) \wedge r$

23 Let

- p: The water is $60^{\circ}\mathrm{F}$.
- q: The sun is up.
- r: We go snorkelling.

Write the following statements in words:

a $(p \wedge q) \vee r$ **b** $(q \implies p) \vee r$

c $(q \wedge r) \implies p$ **d** $\sim p \implies (q \vee r)$

e $q \implies (p \iff r)$